Nagabhushan Deshpande

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Professional Summary

Embedded Systems Engineer with expertise in Embedded C, firmware development, and Real-Time Operating Systems (RTOS). Skilled in implementing secure boot, cryptographic signing, and platform security on TI AM64x. Proficient in ESP32 development using ESP-IDF, device driver development, and communication protocols including UART, SPI, I2C, and ESP-NOW. Strong understanding of bootloaders, memory management, and embedded system architecture.

Education

PES University — Bengaluru, India
Bachelor of Technology – Electronics and Communication Engineering (CGPA: 8.28) — 2021 – 2025

Technical Skills

Programming Languages:	C, C++, Embedded C
Microcontrollers/Processors:	ESP32, STM32, TI AM64x
Operating Systems:	Linux, FreeRTOS , Bare-metal programming
Protocols:	UART, SPI, I2C, ESP-NOW
Security and Architecture:	Bootloaders, Secure boot, Cryptographic signing, Device Drivers
Soft Skills:	Documentation, Problem solving, Collaboration, Git

Work Experience

Celstream Technologies — Oct 2025 – Present
Embedded Software Developer

1. Valve Island Industrial Automation

- Implemented platform security for TI AM64x MPU-based systems, improving system security and that reduces unauthorized access risks by 50%.
- Designed secure boot workflow including key injection, certificate generation, and firmware signing, improving firmware integrity.
- Configured and built secure boot artifacts using TI Secure Keywriter tools and reduced manual configuration by 50%
- Debugged boot issues in bootloader and firmware environments.

Celstream Technologies — Feb 2025 – Oct 2025
Project Trainee-Intern

- Developed firmware using **ESP-IDF** for ESP32 devices that enables low-latency wireless communication using ESP-NOW.
- Configured sensor communication using **UART, SPI, and I2C** protocols.
- Developed applications using FreeRTOS tasks and scheduling mechanisms on ESP32.
- Gained strong understanding of industrial communication protocols, microcontroller architecture, and memory management in embedded systems

Projects

Fire Alarm System

- Designed a fire detection system using **Arduino Uno, a DHT11 temperature sensor, and a MQ-2 gas sensor**, achieving a **detection accuracy improvement of 20%**. Integrated a threshold-based alert mechanism using a buzzer to notify the surrounding environment.
- Demonstrated sensor data acquisition and real-time alert system.

Computer and Communication Networks - Internet Protocol Security

- Simulated a secure network topology using GNS3 with routers and end devices, supporting upto 15+ nodes. Enforced encryption and decryption mechanisms for secure communication over public networks.
- Set up routers and PCs to analyze IPsec-based secure communication using wireshark.

Certifications

Mastering Microcontroller and Embedded Driver Development- Udemy	2026
Computer Science Minor Certificate- PES University	2025
Publication - Certificate of Presentation- IEEE APS	2025